RRISK: Analyzing COVID-19 Risk in Food Establishments

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ABSTRACT

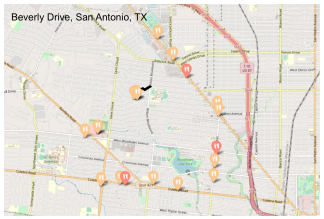
During the COVID-19 pandemic, consumers have faced significant additional health risks and challenges when eating out. In this paper we present RRISK, a comprehensive, visual, and interactive application that allows users to view COVID- 19 risk assessments for restaurants in any area in the United States, search for the safest food establishments in their neighborhood, and find the best places to eat while still minimizing their risk of contracting COVID-19. RRISK provides a color-coded risk assessment for all food establishments in any given area, displaying it on an easy-to-navigate, interactive map platform. We take a data-driven and machine learning-based approach using data collected from appropriate and diverse sources related to COVID-19 risk in the locality, restaurant reviews, customer feedback, and health inspection data, to compute an overall risk score and communicate this information to users. We demonstrate that RRISK has a high accuracy when it comes to modeling restaurant risk, and complements social media ratings for restaurants with more precise risk scores. RRISK has the potential to save lives and completely revolutionize the pandemic dining experience, by connecting the public with the scientific data and analysis needed to make informed, facts-based decisions on if, and where, to eat out during these times.

BACKGROUND AND MOTIVATION

As we enter the 13th month of the pandemic, its impact on our day-to-day lives has been monumental. In the United States alone, there have been 27 million infections and over 460,000 deaths to date. One of the most impacted industries has been the restaurant industry. Fast food and restaurant dining are an integral part of American culture, and before the pandemic more than 56% of Americans ate out at least 2-3 times per week. This rate has remained high even during COVID, and in any given 24-hour period today, 21% of the population will eat out. The significance of this is economic as well as cultural; the restaurant industry is the second-largest job sector in the US, and employs a disproportionately high percentage of racial minorities, who are also at a far elevated risk of catching the virus.

On the consumer side, studies have also shown that the commonly advised safety precautions, like wearing masks, don't actually help much when it comes to restaurant dining; the risk of catching coronavirus at a restaurant is correlated more with the restaurant itself than the level of one's personal caution, and those who dine out with versus without masks seem to catch the virus at similar rates.

So, what are we left with? We have a population that remains dependent on eating out, and a situation where simply exercising personal caution by wearing masks and social distancing isn't nearly enough. The focus then shifts to a much bigger question: how can we help consumers dine out in a safe and informed way?



RRISK OVERVIEW

RRISK is a comprehensive, visual, and interactive web tool that allows users to view live COVID-19 risk assessments for any bar or restaurant in the US, search for the safest food establishments in their jurisdiction, and find the best places to eat while still minimizing their risk of contracting COVID-19. RRISK provides color-coded risk assessment for all food establishments in a given area, and displays it on an easy-to-navigate interactive map platform. The user simply enters his/her location and a search radius, and the system returns all the restaurants and bars in the target area displayed, with dynamically calculated risk statistics and color-coded labels. RRISK gleans data from diverse sources and combines them in a machine learning algorithm in a Python framework to feed the application. The algorithm takes a detailed and systematic approach to computing contraction risk by analyzing a variety of factors from a range of up-to-date and reliable data sources, including population density, case counts and growth rate, restaurant ratings and written reviews, health inspection data, and restaurant categories and tags. A screenshot of part of the RRISK map view for San Antonio, Texas, is shown at the bottom of the first column. One can see that within a small radius of just 12 miles, there is a large diversity of COVID-related dining risks at different restaurant

DATA

Our data spans seven dimensions and scales to 11.5 million rows of information, and the factors considered are shown below. At the local and municipal level, consider population demographics, as well as COVID-19 case and growth data. At the establishment level, we take the Yelp rating of every restaurant, as well as the top three most helpful recent reviews for that restaurant, and its food/restaurant tags. We apply text mining techniques to the review data and extract information about the quality and cleanliness of the restaurant, based on identified keywords. Similarly, we extract 42 restaurant types, for example "bistro" or "juice bar", from over 192 Yelp tags mined, giving us additional information about cuisine type, which allows us to provide safety information at the tag level.

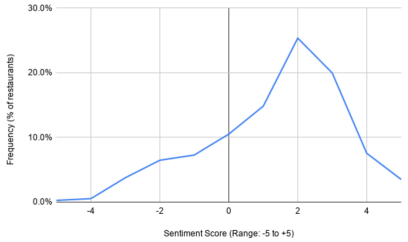
Type	Level	Data Source	Data size
Population Density	City	US Census Bureau	5000 cities with population over 5k
Case count	County	New York Times	420K rows
Case growth rate	County	New York Times	420K rows
Restaurant ratings	Business	Yelp	650k restaurants across US
Restaurant reviews	Business	Yelp	3.25M reviews
Health Inspection data	Business	Data.gov	6.5M rows
Restaurant type	Business	Yelp	192 tags reduced to 42 of interest

ALGORITHM

RRISK first takes the case total per county and the population of the county and use it to compute cases per 100k residents. We use a data-driven approach to scaling these values. For counties with high case rates, we give a near-zero weight, as higher scores are considered favorable and lower scores are considered unfavorable, and logarithmically scale the remainder of the weights based on occurrence rates. We use the total cases and cases 10 days prior to compute case growth rates, and take a similar approach and normalize that score. We also create scaled scores for population density and average restaurant rating.

Another factor we take into account is restaurant tags that tell us more about the restaurant. Here we identify 192 such tags, of which 42 are of particular interest to us. We use this in conjunction with the idea that the less interaction a cook or employee has with the food itself, the safer it is to eat. These risk weights are based on early CDC recommendations, since while evidence does not clearly point to SARS-CoV-2 as a food-borne virus, opportunities for an infected chef or employee to transmit the virus to the plate or other surface, vary widely by the method of food preparation.

For example, buying pre-packaged food is much less risky than going to a gourmet restaurant, because the interaction of the chefs with the food is much less, and so there is less COVID transmission opportunity. As a result, tags such as sushi, which is hand-prepared and uncooked, get a negative score, while tags such as soup, which generally has a simple preparation protocol and limited contact with chefs, earn a positive score. Additionally, establishment type is also relevant; take-out or takeaway foods, coffee shops, bentos, and bakeries are likely the to be the least risky, while those such as bars, beer halls and hookah bars, food courts and buffets are most risky. We tag about 42 restaurant types mined from Yelp tags, and incorporate them into our risk computation algorithm.



We also consider sentiment analysis of written comments. Written reviews of a restaurant are generally a good way to gauge overall quality, since people tend to be reliable in voicing concerns if something is seriously amiss. Comments about poor sanitation and hygiene imply that the establishment is probably not following best cleanliness practices, and most likely is not safe, especially in the time of COVID where we have to be extra cautious in these regards. We parse the comments for each restaurant, and mine for both positive and negative keywords related to cleanliness, then generate a frequency-based comment score. For example phrases like "not clean" or "smelled bad" or "unsanitary" would yield a poor score, but phrases like "were all wearing masks" would yield a good score. The comment score is based on the rated top 5 most helpful reviews, since this ensures consistency in standards between restaurants with many reviews and those less frequently visited ones. The distribution of sentiment scores is shown above; the score range is from -5 to +5, and despite a slight skew to the left, we can still see a healthy and diverse distribution of scores. also helps to ensure formula consistency across both rural and urban areas. Factor analysis determined that restaurant reviews, especially those that discuss sentiments beyond just the taste of the food, are useful in assessing COVID-19 risk and significantly improved the accuracy of our algorithm. Finally, local health inspection data is also considered.

TRAINING, VALIDATION AND VERIFICATION

In order to train our model, we used a set of 500 randomly selected restaurants representative of the national sample, for which parameters are known. Ground truth values corresponding to the various color bands were created by 10 human volunteers, who were asked to review every restaurant in order to reduce bias. We then applied regression to determine the weights and scaling that would go into each factor, in order to create a formula for the final RRISK score. Cutoff values for each color category were predetermined, and the ranking algorithm was modified to adhere to placement of establishments at scores falling within these categories based on our ground truth data. Validation and testing of our model was completed using another 250 randomly selected restaurants, and truth values for these were determined by a second group of 10 human volunteers. Our model was found to be accurate for 87% (217 out of 250) of the cases.

We also tracked a number of restaurants nationwide that were shutdown by health authorities for COVID-related violations, to see if our system predicted such an occurrence before the shut down. We measured this by taking a random sample of food establishments around the United States that were closed down. For these establishments, the average Yelp rating was found to be 4.2, which would generally indicate a very good restaurant, and gives no indication to the user that these restaurants were in fact violating health and hygiene rules, and were so high-risk for customers that they had to be forcibly closed down by the state. However, the RRISK algorithm was able to measure some level of COVID-19 risk for over 90% of these restaurants that were closed down, and for 70% of them flagged them as high risk or worse, further strengthening its validity as a reliable model of COVID-19 risk prediction.

INTERESTING INSIGHTS

Risky region with a reasonable restaurant



Same cuisine, same location, different risk



The first image, on the left, shows a number of restaurants along a street in Rockford, Illinois. From the color-coding trends, we see that this area is generally high-risk for COVID-19. However, one restaurant, the Trop Club, has an orange tag, showing a noticeably lower risk level than the rest of the area. In fact, the Trop Club has positive customer reviews, a 5-star Yelp rating, and documented strict adherence to COVID-related safety procedures. This provides another example of how the RRISK algorithm is able to take multiple risk factors into account, and even in areas which are considered to be generally high-risk based on the coronavirus case data, identify those establishments that are doing everything they can to ensure customer safety as the safest options in the area.

The second image, on the right, shows two restaurants in Yonkers, New York. These restaurants are quite similar: both serve Cuban cuisine, are located in similar areas mean there is an identical prevalence of the virus in both places, and their front doors are less than 200m apart. Yet, one has a low-risk score of 63, while the other has a very high-risk score of 46. These assessments are consistent with local health inspection findings as well as user reviews, highlighting the significant risk variation across otherwise similar restaurants. The consistency with local health inspection data and written reviews despite near-identical locations for each restaurant also provides an example of the high levels of accuracy and reliability our algorithm provides.

CONCLUSION

We have demonstrated RRISK to be a highly reliable and much-needed data-driven assessment tool for COVID-19 risk in food establishments, that allows users to make the safest possible dining choices during the pandemic. As next steps, we plan to expand our system to international data, and also create a generalized framework that can be applied to other public health issues in the future.

REFERENCES:

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